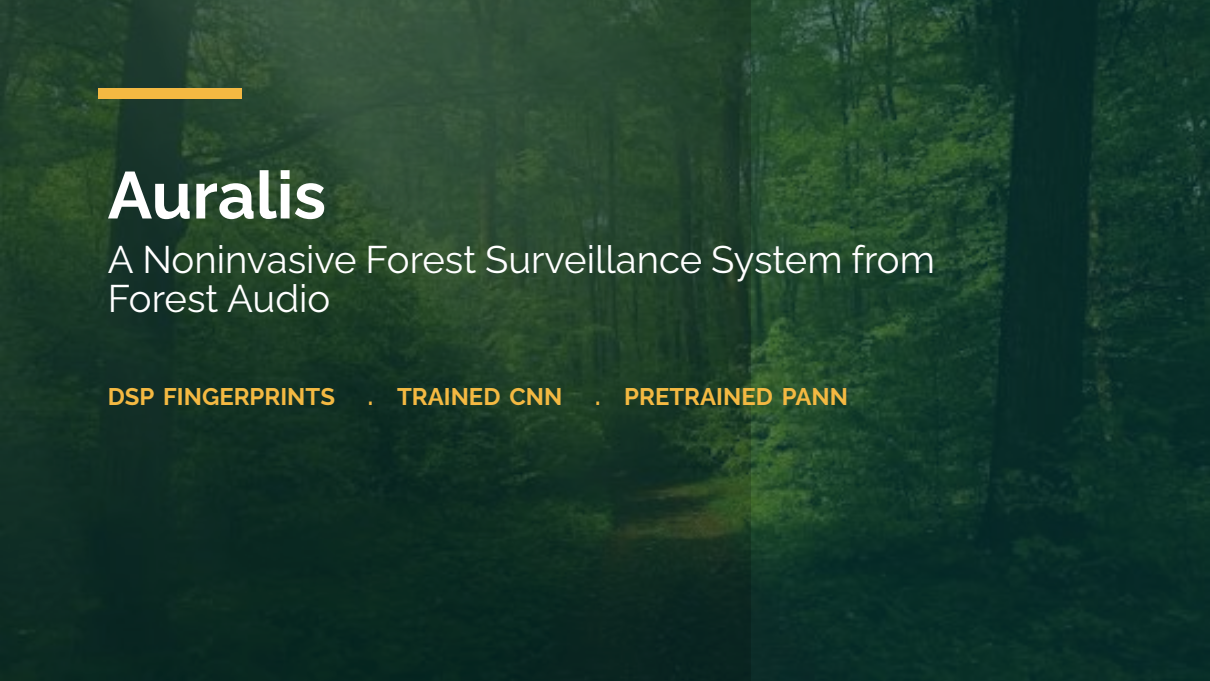




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# Auralis

A Noninvasive Forest Surveillance System from  
Forest Audio

DSP FINGERPRINTS . TRAINED CNN . PRETRAINED PANN

# Problem statement



## Input

A forest audio recording of arbitrary duration, format, sample rate, and recording quality. An optional site/location tag restricts the candidate species list; it is a categorical filter, not a continuous model feature.

## Output

A confidence for every candidate species, thresholded into the final list of species judged present.

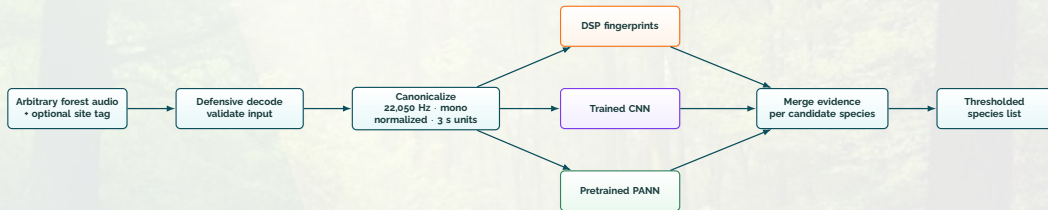
## Signals-lab constraint

The team implements resampling, normalization, filtering, windowing, and spectrogram generation rather than outsourcing the DSP chain to black-box libraries.

## Reliability requirement

The pipeline must safely accept unfamiliar formats, rates, durations, and quality levels. Species outside the trained label space are a model-coverage limitation, not an ingestion failure.

# Proposed solution



## Why three paths?

DSP finds recurring patterns, the project CNN learns target classes, and PANN contributes broad pretrained knowledge. Their different failure modes strengthen the ensemble.

## Site-aware filtering

Location removes biologically impossible candidates before final ranking. It never substitutes for acoustic evidence or enters the model as a numeric feature.

# Canonical audio contract



## Every processing unit

|                        |                        |
|------------------------|------------------------|
| <b>Sample rate</b>     | 22,050 Hz              |
| <b>Channels</b>        | Mono; average down-mix |
| <b>Loudness</b>        | Fixed reference level  |
| <b>Duration</b>        | 3 seconds              |
| <b>Native formats</b>  | WAV and FLAC           |
| <b>Optional format</b> | MP3 via decoder        |

## One boundary

Training and uploaded audio use the same canonicalization. Long files become overlapping 3-second units; short units are handled consistently.

## Why it matters

Fixed rate, channel count, scale, and duration prevent a classifier from mistaking accidental input differences for biological evidence.

## Documented exceptions

MP3 uses a standard call only for container/codec parsing; lossy compression may already have removed high-frequency evidence. PANN resamples its canonical copy to its model-required 32 kHz.

## Robust audio handling

- ▶ WAV/FLAC accepted natively; MP3 decoding optional.
- ▶ Team-built resampling, downmixing, normalization, filtering, windowing, and spectrograms.
- ▶ Sliding 3-second units support arbitrary recording lengths.
- ▶ Unfamiliar rates and quality levels are canonicalized before inference.

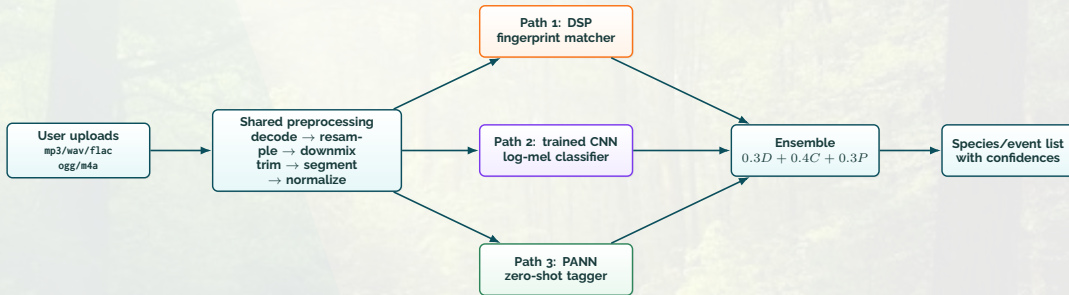
## Detection behavior

- ▶ Confidence per candidate species, followed by thresholding.
- ▶ Optional location filtering narrows biological candidates.
- ▶ DSP, CNN, and PANN failures are isolated and reported rather than crashing the full analysis.
- ▶ Multi-path evidence supports species and relevant acoustic-event detection.

## Scope boundary

Robust ingestion does not imply recognition of a species absent from every model's label space. Extending biological coverage is a data/model task.

# The big picture



|                          | DSP                                              | CNN                                                         | PANN                                         |
|--------------------------|--------------------------------------------------|-------------------------------------------------------------|----------------------------------------------|
| <b>Nature</b>            | Hand-built signal matching                       | Supervised model trained from scratch                       | Pretrained CNN14/AudioSet                    |
| <b>Training Question</b> | References only<br>"Does this pattern reappear?" | 50 epochs on synthetic data<br>"Which class looks similar?" | No fine-tuning<br>"What does AudioSet know?" |

# Shared preprocessing: one upload, consumer-specific outputs



## AudioPreprocessor

Driven by a ConsumerSpec: target sample rate, channels, window duration/hop, and amplitude range. One raw upload is forked into the exact representation each path needs.

- ▶ Trim at 35 dB below peak; keep internal pauses.
- ▶ DSP receives the whole clip; neural paths receive overlapping windows.
- ▶ Peak normalization targets each consumer's expected range.

## Shared convolution engine

FIR resampling and bandpass filtering use the convolution theorem:

$$x * h = \text{IFFT}(\text{FFT}(x) \text{FFT}(h)).$$

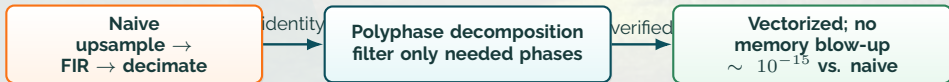
Decoding tries soundfile first and uses librosa.load as the fallback for compressed containers.

## Original failure

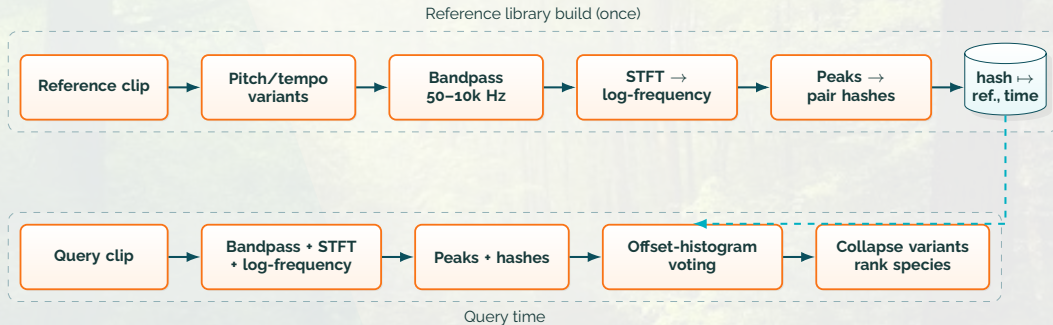
A windowed-sinc low-pass FIR supplies anti-aliasing. Naively upsampling by  $L$ , filtering, then decimating by  $M$  creates an intermediate array  $L$  times longer than the input. For  $22,050 \rightarrow 48,000$  Hz,  $L = 320$ —large enough to trigger an out-of-memory kill on a real clip.

## Polyphase fix

1. Reduce  $L/M$  with Fraction; hence  $\gcd(L, M) = 1$ .
2. Split the FIR into  $L$  short phase filters.
3. Convolve phases directly with the original signal.
4. Fold decimation into phase selection using  $M^{-1} \bmod L$ .



# Pathway 1: DSP fingerprint matcher



Inspired by acoustic fingerprinting: identify reproducible constellations of spectrogram peaks, then demand temporal consistency rather than mere spectral similarity.

# DSP representation: bandpass, STFT, and log frequency



## Bandpass

50–10,000 Hz, constructed as the difference between high-cut and low-cut low-pass FIRs. It reuses the resampler's FIR design and FFT convolution.

## STFT

46 ms Hann windows, 12 ms hop (about 74% overlap). Longer windows sharpen frequency but blur time. Hann tapering reduces edge-discontinuity leakage.

## Log representation

$20 \log_{10} |X|$  prevents a few loud bins dominating. A 128-bin triangular,  $\log_2$ -spaced filterbank makes pitch multiplication become an additive vertical shift.

## Resolution caveat

Zero-padding the FFT creates smoother interpolation, not new frequency resolution; true resolution is fixed by the analysis-window duration.

## Peak picking

`maximum_filter` finds local maxima in frequency-time neighborhoods. A peak survives only if it passes:

- ▶ global floor: within 40 dB of the clip maximum;
- ▶ adaptive row floor: 15 dB above that row's median;
- ▶ zoning cap: limited peaks per frequency-time region.

The adaptive floor resists steady narrowband interference; zoning preserves quieter call segments.

## Pair hashing

Each anchor connects to up to `fan_out` later peaks inside a target-time zone. Pairs are distinctive; individual peaks collide too often.

$$h = \text{pack}(f_1, f_2, \Delta t).$$

Pitch-invariant mode instead uses

$$h = \text{pack}(f_2 - f_1, \lfloor \Delta t / 2 \rfloor),$$

trading extra collisions for pitch and mild stretch tolerance.

# DSP matching: offset-histogram voting



For every shared hash and reference occurrence,

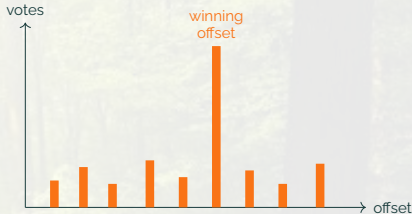
$$\Delta = t_{\text{query}} - t_{\text{reference}}.$$

The matcher counts votes by (reference,  $\Delta$ ). A true occurrence aligns many hashes at one offset; accidental collisions scatter across offsets.

## Score

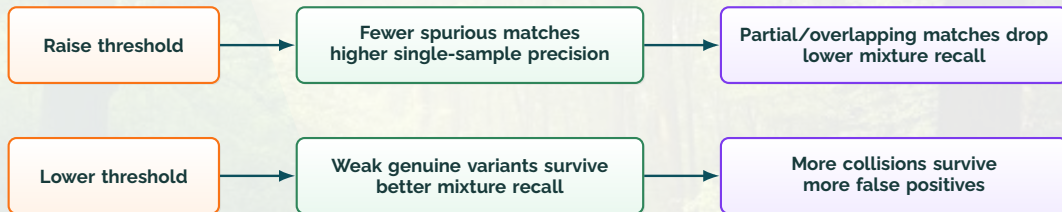
$$\text{score} = \frac{\text{votes at winning offset}}{\text{reference's total hash count}}.$$

It is an uncalibrated overlap fraction, not a probability.



Raw shared-hash count alone is insufficient: temporal agreement is the discriminating signal.

# The central DSP trade-off: min\_raw\_count



## Production decision

MIN\_RAW\_MATCH\_COUNT = 5: deliberately low so pitch/tempo variants and partial events survive long enough to vote. The choice favors multi-event robustness over maximum single-clip precision.

## Reference variants

Each reference expands over  $\pm 2$  semitones and rates 1.0/1.15 (six pitch-tempo variants). Variants score independently; only afterward does `collapse_variant_scores` retain the best variant per species. Premature merging would dilute a match that resembles one variant.

## Window aggregation

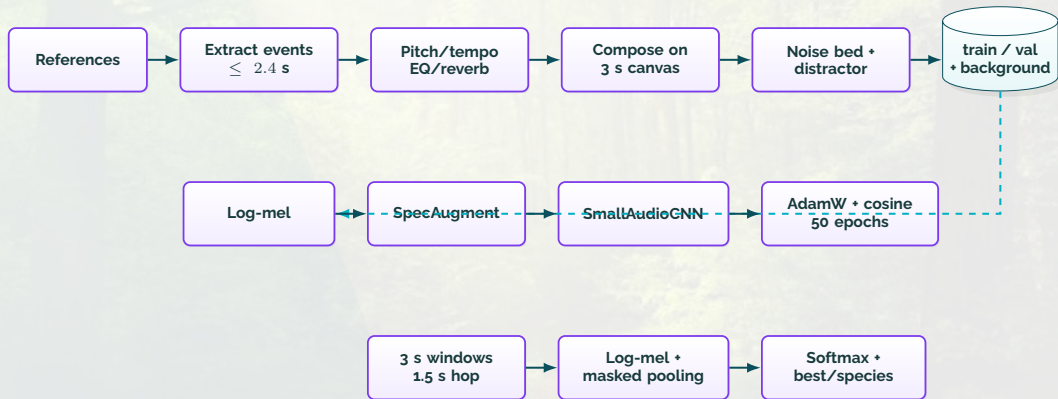
Keep the maximum per species across query windows and stored references. Averaging would penalize a real call that resembles only one reference or occurs in only one window.

## `dsp_diagnostics.py`

- ▶ flags suspiciously low reference hash counts;
- ▶ reports hashes per second to separate short clips from corrupt inputs;
- ▶ explains occasional self-match scores above 1: repeated query hashes create an  $n \times m$  cross-product against self-similar references.

**Output:** ranked {species, confidence} at 22,050 Hz mono; confidence is uncalibrated fingerprint overlap.

## Pathway 2: trained phase-1 CNN



# Why simple crop-or-pad failed



## Shortcut learned by the model

References span 0.6–8 s. Zero-padding a 0.6 s bark to 3 s makes about 80% of the input digital silence, while a long crow clip fills the canvas. The CNN can identify “amount of silence” instead of the call.

## Measured evidence

The same test sounds scored 88% with zero-padding but only 48% after replacing silence with a realistic noise floor; cat, cow, and dog fell to 0%. The drop exposed leakage, not a deployment-safe model.

Trim + split events  
merge gaps < 0.3 s

Augment actual event  
not padding

Place randomly; edge cuts;  
repeat short calls 1–3×

Fill canvas with realistic  
background at 3–35 dB SNR

## Balanced event composition

- ▶ Each class receives exactly `-variants-per-class` clips (typically 200), independent of source duration.
- ▶ Event transforms: pitch  $\pm 3$  st, tempo  $0.8\text{--}1.25\times$ , band-limiting EQ, and synthetic reverb.
- ▶ Random location, occasional edge truncation, call repetition, and optional quiet call from another species.
- ▶ `_background` uses the same generator, so “nothing” belongs to the same acoustic world.

## noise\_bed.py

White, pink ( $1/\sqrt{f}$ ), brown, mains hum with harmonics, amplitude-modulated low-pass wind, and near-silent quiet. Beds are normalized to unit RMS, then scaled to the requested SNR.

## No tiling

Repeating short audio produces autocorrelation near 0.9 versus roughly 0 for real audio—an artificial periodic shortcut. Tiling was rejected in both noise generation and dataset composition.

## Input representation

3 s at 22,050 Hz  $\rightarrow$  64-bin log-mel spectrogram using  $n_{\text{fft}} = 1024$ , hop 256, and 50–10,000 Hz; fixed frame count and scaling approximately to  $[-1, 1]$ .

## SmallAudioCNN

Conv(1 $\rightarrow$ 16)-BN-ReLU-Pool  
 $\rightarrow$  Conv(16 $\rightarrow$ 32)-BN-ReLU-Pool  
 $\rightarrow$  Conv(32 $\rightarrow$ 64)-BN-ReLU  
 $\rightarrow$  masked global average pool  
 $\rightarrow$  Dropout(0.3)  $\rightarrow$  Linear(64 $\rightarrow$ classes)

## Masked pooling

Short uploads require noise-bed padding. Convolutions and BatchNorm still process it, but `valid_frames` gives padded columns exactly zero weight in global average pooling. This is a structural guarantee, not a statistical hope.

Training samples are unmasked because their entire 3 s canvas is intentionally generated content plus realistic background. The network remains small because roughly one real recording per class, even heavily augmented, cannot support a large model without rapid overfitting.

## Training recipe

|                |                               |
|----------------|-------------------------------|
| Epochs         | 50                            |
| Optimizer      | AdamW; weight decay $10^{-2}$ |
| Schedule       | Cosine annealing              |
| Loss           | Cross-entropy; smoothing 0.1  |
| Batch / LR     | 16 / $10^{-3}$                |
| Regularization | Dropout + SpecAugment         |
| Validation     | 15%; best accuracy, then loss |

## Honest validation

With at least three recordings per class, whole recordings are held out. With fewer, train and validation use augmentations of the same recordings; the resulting optimistic estimate is detected and logged per class. More independent real recordings are the largest accuracy lever.

## Inference

Up to 120 s; 3 s windows with 1.5 s hop; best probability per species. A background winner or top score below 0.20 returns no confident detection, preventing closed-set softmax from always inventing a class.

## Pathway 3: PANN zero-shot tagging



### What it contributes

CNN14 was pretrained on roughly two million AudioSet clips spanning 527 general sound classes. It is used without fine-tuning, providing broad prior knowledge independent of Auralis references and synthetic training.

Zero-padding of short trailing windows may soften scores because CNN14 exposes no masking hook; this is accepted for the exploratory pathway. Sigmoid output is genuinely multi-label.

### Documented exception

The pretrained mel front end is inseparable from the learned weights; replacing it would defeat the purpose of transfer. PANN is therefore an explicitly exploratory black-box path, while its required 32 kHz resampling remains hand-built.

## Why mapping is necessary

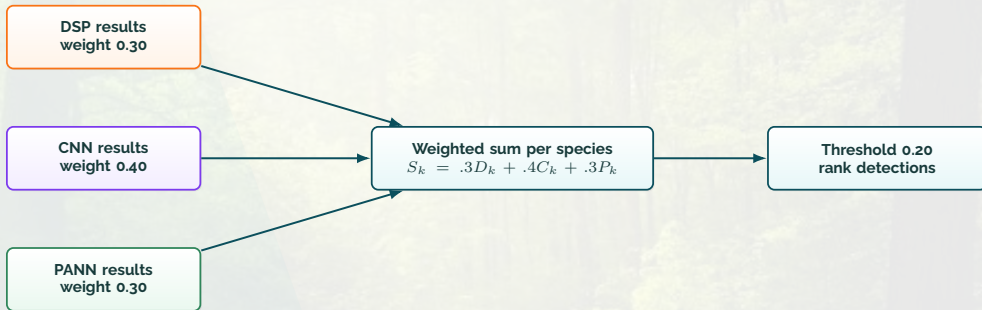
Raw AudioSet labels are both redundant (meow, cat, caterwaul) and out of scope (e.g., rain or music). They cannot be treated as species predictions or merged directly with the other paths.

## Mapping rules

- ▶ Synonyms map to one category: bark/dog/bow-wow → dog.
- ▶ Combine synonyms by maximum, never sum, to avoid correlated inflation.
- ▶ Use `farm_animals` only if no specific farm species clears threshold.
- ▶ Suppress generic `animal/bird/noise` tags once a specific category fires; retain as fallback otherwise.
- ▶ Preserve unmapped tags for diagnostics, but exclude them from the ensemble.

Output: relabeled, ranked multi-label detections; minimum PANN score 0.15.

# Ensemble: independent failures, shared decision



## Robust execution

`process_ensemble_sample()` gives CNN and PANN separate temporary-file copies while DSP reads independently, preventing one consumer from leaving another at end-of-file. Every path is called defensively.

## Common contract

All paths return `{species, confidence}` and apply the common 0.20 backend threshold. PANN is restricted to mapped Auralis categories. Missing checkpoints or other failures appear in `path_errors`; remaining paths still return.

The backend returns per-path predictions; the client applies the displayed weights.

# Reproducible generated test set



## Eight stress variants per species

Original; padded silence; white noise at 10 dB SNR; pitch  $+3/ - 3$  st; speed  $0.8\times/1.25\times$ ; and a fixed  $-12$  dB, 1 kHz tone to exercise the adaptive row floor.

## Mixtures

Five non-overlapping concatenations and five randomized overlapping mixtures test multi-event aggregation. The first mixture of each kind always includes gunshot.

## Ground truth, not filename guesses

`generate_test_samples.py` uses a fixed seed. `generated_manifest.json` records source, transform, and exact event timestamps for every mixture.

## Per-path harnesses

`test_dsp.py`, `test_cnn.py`, and `test_pann.py` cache raw JSON and chart single-event top-1 plus mixture confidence/species recovery, separating overlap regimes.

# Evaluation: defensible conclusions



| Path | Observed qualitative pattern                                                                                            | Expected limitation                                                                                                               |
|------|-------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| DSP  | Strong on clean/original clips (often 0.8–1.0); moderate pitch, tempo, and noise tolerance from invariants and variants | Heavy transforms can produce no detection; mixtures often recover only a subset, consistent with the raw-vote threshold trade-off |
| CNN  | Generally confident top-1 across most generated variants (often $\geq 0.8$ )                                            | Harder transforms may fall below threshold; validation is optimistic when independent recordings are unavailable                  |
| PANN | Recognizes several species with no project-specific training                                                            | Misses or weakens out-of-distribution calls and project mixtures; labels require careful mapping                                  |

## Reporting policy

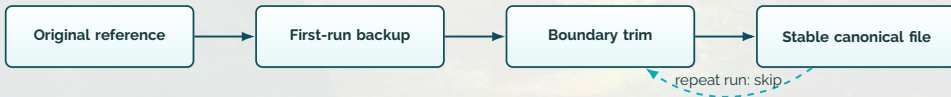
Exact accuracy changes when `samples/test/` is regenerated. Presenting the fixed-seed charts and qualitative cross-path pattern is more defensible than one unstable headline number.

## `trim_references.py`

Trims only leading/trailing silence in place and backs up the untouched original on the first run. Later runs are no-ops, preventing compounding lossy re-encoding.

## Container handling

Because the writer cannot encode M4A, trimmed M4A references become WAV and the original moves to backup. The dataset builder therefore cannot double-count both versions.



# What is technically distinctive



- ▶ Polyphase identity fixes a real OOM without replacing the hand-built resampler.
- ▶ Log-frequency relative hashing converts pitch scaling into translation invariance.
- ▶ Offset voting distinguishes coherent occurrences from coincidental collisions.
- ▶ The `min_raw_count` setting exposes an explicit precision-mixture-recall decision.
- ▶ Masked pooling gives padding provably zero classifier weight.
- ▶ The 88% to 48% experiment uncovered silence leakage and motivated realistic composition.
- ▶ Tiling was independently rejected in two subsystems because of artificial periodicity.
- ▶ Generated tests carry exact, reproducible per-event ground truth.
- ▶ Validation labels itself optimistic when recording-level holdout is impossible.
- ▶ PANN's black-box boundary and required relabeling are explicit, while resampling stays hand-built.

# One-page comparison



| Aspect                  | DSP                                              | CNN                                         | PANN                                      |
|-------------------------|--------------------------------------------------|---------------------------------------------|-------------------------------------------|
| Core idea               | Peak-pair hashing + offset votes                 | Learned log-mel classifier                  | Pretrained CNN14; relabeled               |
| Training Representation | References only<br>Bandpassed log-frequency STFT | Synthetic data; 50 epochs<br>64-bin log-mel | None<br>Internal pretrained mel front end |
| Invariance              | Relative pitch, mild tempo, variant grid         | Augmentation + SpecAugment                  | AudioSet coverage                         |
| Multi-event             | Best/species votes/windows over                  | 3 s sliding windows                         | 6 s multi-label windows                   |
| Main dial               | min_raw_count                                    | confidence/background threshold             | confidence + label mapping                |
| Output Weight           | Uncalibrated overlap<br>0.30                     | Softmax probability<br>0.40                 | Relabeled sigmoid<br>0.30                 |

## Bottom line

The three paths fail differently: exact-pattern evidence, project-specific learned similarity, and broad pretrained knowledge. Shared preprocessing makes them comparable; defensive execution and a common vocabulary make them mergeable.

## Appendix: input/output and tunables



|                       | DSP                                                                       | CNN                                                         | PANN                                      |
|-----------------------|---------------------------------------------------------------------------|-------------------------------------------------------------|-------------------------------------------|
| <b>Input limit</b>    | One uploaded file / whole clip                                            | Up to 120 s                                                 | Up to 120 s                               |
| <b>Preprocess</b>     | 22,050 Hz mono; 50–10k; 46/12 ms; 128 log bins                            | 22,050 Hz; 3/1.5 s; 64 mels                                 | 32,000 Hz; 6/3 s                          |
| <b>Reference/data</b> | samples/references/*; six variants                                        | 200/class + _background                                     | AudioSet pretraining                      |
| <b>Threshold</b>      | common 0.20 after DSP scoring                                             | MIN_CONFIDENCE=0.20                                         | internal 0.15; common 0.20 for ensemble   |
| <b>Key controls</b>   | fan-out, target $\Delta t$ , pitch invariance, time bucket, minimum votes | augmentation, SpecAugment, masked pooling, background class | mapping, fallback suppression, confidence |

# Samples for Benchmark



## Single-event: 88

cat\_noised.wav  
cat\_original.wav  
cat\_padded\_silence.wav  
cat\_pitch\_decreased.wav  
cat\_pitch\_increased.wav  
cat\_slowed.wav  
cat\_speeded.wav  
cat\_toned.wav  
cow\_noised.wav  
cow\_original.wav  
cow\_padded\_silence.wav  
cow\_pitch\_decreased.wav  
cow\_pitch\_increased.wav  
cow\_slowed.wav  
cow\_speeded.wav  
cow\_toned.wav  
crow\_noised.wav  
crow\_original.wav  
crow\_padded\_silence.wav  
crow\_pitch\_decreased.wav  
crow\_pitch\_increased.wav  
crow\_slowed.wav  
crow\_speeded.wav  
crow\_toned.wav  
dog\_noised.wav

dog\_original.wav  
dog\_padded\_silence.wav  
dog\_pitch\_decreased.wav  
dog\_pitch\_increased.wav  
dog\_slowed.wav  
dog\_speeded.wav  
dog\_toned.wav  
frog\_noised.wav  
frog\_original.wav  
frog\_padded\_silence.wav  
frog\_pitch\_decreased.wav  
frog\_pitch\_increased.wav  
frog\_slowed.wav  
frog\_speeded.wav  
frog\_toned.wav  
goat\_noised.wav  
goat\_original.wav  
goat\_padded\_silence.wav  
goat\_pitch\_decreased.wav  
goat\_pitch\_increased.wav  
goat\_slowed.wav  
goat\_speeded.wav  
goat\_toned.wav  
gunshot\_noised.wav  
gunshot\_original.wav

## Non-overlapping: 5

gunshot\_padded\_silence.wav  
gunshot\_pitch\_decreased.wav  
gunshot\_pitch\_increased.wav  
gunshot\_slowed.wav  
gunshot\_speeded.wav  
gunshot\_toned.wav  
horse\_noised.wav  
horse\_original.wav  
horse\_padded\_silence.wav  
horse\_pitch\_decreased.wav  
horse\_pitch\_increased.wav  
horse\_slowed.wav  
horse\_speeded.wav  
horse\_toned.wav  
owl\_noised.wav  
owl\_original.wav  
owl\_padded\_silence.wav  
owl\_pitch\_decreased.wav  
owl\_pitch\_increased.wav  
owl\_slowed.wav  
owl\_speeded.wav  
owl\_toned.wav  
roaring-cat\_noised.wav  
roaring-cat\_original.wav  
roaring-cat\_padded\_silence.wav

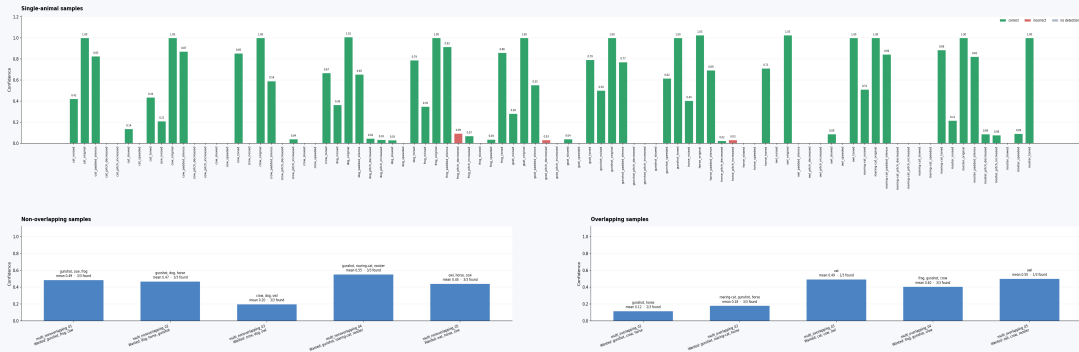
## Overlapping: 5

roaring-cat\_pitch\_decreased.wav  
roaring-cat\_pitch\_increased.wav  
roaring-cat\_slowed.wav  
roaring-cat\_speeded.wav  
roaring-cat\_toned.wav  
rooster\_noised.wav  
rooster\_original.wav  
rooster\_padded\_silence.wav  
rooster\_pitch\_decreased.wav  
rooster\_pitch\_increased.wav  
rooster\_slowed.wav  
rooster\_speeded.wav  
rooster\_toned.wav  
multi\_nonoverlapping\_01.wav  
multi\_nonoverlapping\_02.wav  
multi\_nonoverlapping\_03.wav  
multi\_nonoverlapping\_04.wav  
multi\_nonoverlapping\_05.wav  
multi\_overlapping\_01.wav  
multi\_overlapping\_02.wav  
multi\_overlapping\_03.wav  
multi\_overlapping\_04.wav  
multi\_overlapping\_05.wav

# Benchmark: DSP Fingerprint Matcher



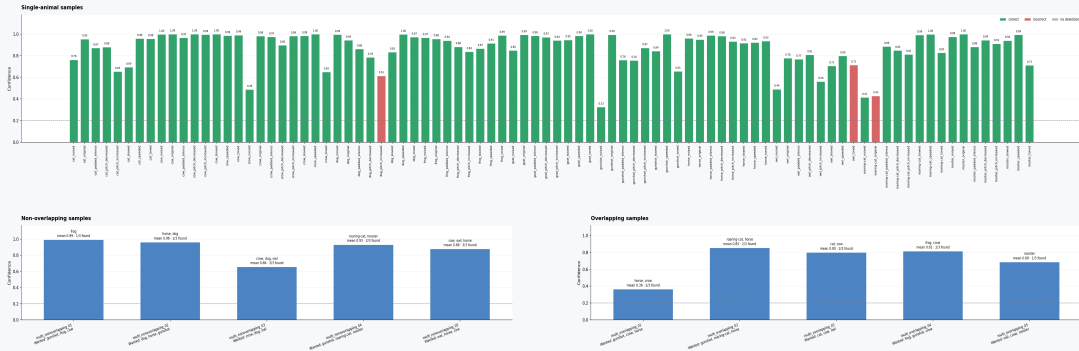
Auralis DSP results

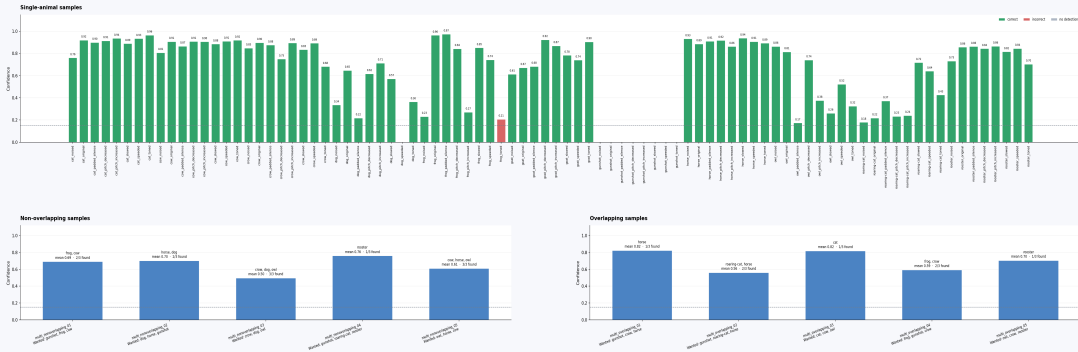


# Benchmark: Trained CNN



Auralis CNN results





# Auralis

Three complementary signal analysis techniques. One decision.

DSP FINGERPRINTS · TRAINED CNN · PRETRAINED PANN

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